

PROJECT DESIGN PHASE

Solution Architecture

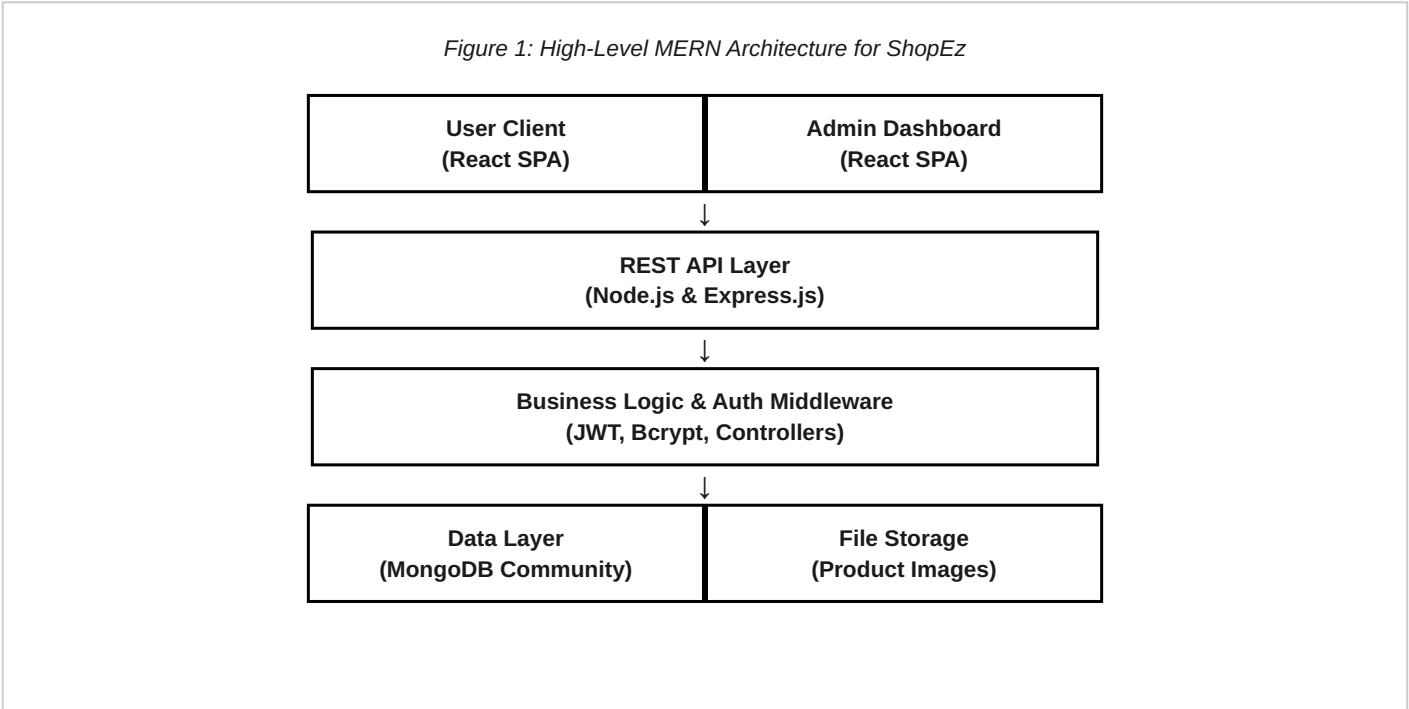
Date	18 June 2026
Team ID	
Project Name	ShopEz (Full Stack MERN-Based E-Commerce)
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a critical engineering process that bridges technical implementation with business objectives. For ShopEz, the architecture is designed to handle scalable product management and secure transactions using the MERN stack.

- Leverage the MERN stack to solve e-commerce navigation and management challenges.
- Provide a structured schema for Users, Products, and Orders using MongoDB.
- Ensure secure API communication through Express.js and JWT authentication.
- Deliver high-responsiveness via a React-based single-page application (SPA).

ShopEz - Solution Architecture Diagram:



Component Descriptions:

- Client Layer (React):** Two distinct frontends providing intuitive navigation for customers and specialized CRUD controls for site administrators via component-based UI.
- API & Server Layer (Node/Express):** Handles secure HTTP requests, input validation, and routes management, ensuring high availability and seamless data delivery through asynchronous logic.
- Data & Storage Layer (MongoDB):** Uses NoSQL collections (Users, Products, Orders) to store flexible document structures, coupled with local image storage for product assets.
- Logical Data Flow:** User interactions trigger API calls; the server validates sessions via JWT, queries MongoDB, and returns JSON responses displayed dynamically by React.